Week 10 Lab Report

Machine Learning & Dashboards: Predicting Student Scores
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1. Introduction

This report presents the work done in the Week 10 lab. The goal was to use machine learning to predict a student's final score from their quiz average, assignments average and attendance, to identify students at risk of failing, and to share the findings in a simple dashboard. All models were evaluated on students hidden from the model during training.

2. Data and Method

The dataset *week10_students.csv* contains 150 clean student records with the columns student_id, name, gender, program, district, attendance_pct, quiz_avg, assignments_avg and score (the target). The three numeric features were used as inputs (X) and score as the target (y). The data was split with `train_test_split(test_size=0.25, random_state=42)`, giving 112 training and 38 test students. The test students were hidden so the model could be checked on students it had never seen, which avoids over-optimistic results from memorisation. Python, pandas, scikit-learn and Plotly were used.

3. Results

3.1 Exploring the data

The correlation of score with each feature shows that quiz average is the most useful predictor.

Feature	Correlation with score
quiz_avg	0.80
assignments_avg	0.54
attendance_pct	0.24

3.2 Regression: predicting the score

A linear regression was trained on the 112 training students. Holding the other features fixed, one extra quiz point adds about 0.58 points to the predicted score.

Term	Coefficient
quiz_avg	0.579
assignments_avg	0.365
attendance_pct	0.113
intercept	-4.428

On the 38 unseen test students the model achieved **MAE = 3.33 points** and **R^2 = 0.89**. In plain language: predictions are off by about 3 points out of 100 on average, and the three inputs explain about 89% of the differences between students' scores. For a new student with quiz 70, assignments

75 and attendance 90, the model predicts a score of 73.7. An invented student (quiz 65, assignments 80, attendance 95) is predicted to score 73.2.

3.3 Classification: catching at-risk students

Students with a score of 50 or more were labelled as passed. In the data, 131 students passed and 19 failed. A logistic regression was trained to predict pass or fail.

Measure	Value
Model accuracy (test set)	0.895
Lazy baseline (predict pass for everyone)	0.895

Confusion matrix	Predicted fail	Predicted pass
Actually failed (4 students)	3	1
Actually passed (34 students)	3	31

The model and the lazy baseline have identical accuracy because most students pass. However, the lazy model never predicts a fail, so it finds none of the failing students. Our model correctly flags 3 of the 4 failing students in the test set (with 3 false alarms and 1 miss). Accuracy alone hides this, so the confusion matrix must be read for imbalanced classes.

3.4 Feature engineering

Three versions of the regression were compared on the same test set.

Features used	Test R^2
1. quiz_avg only	0.624
2. three numeric features	0.887
3. three numeric features + program (one-hot)	0.880

Adding assignments and attendance improved the model strongly (0.62 to 0.89). Adding the program did not help (0.88). More features are not automatically better; they must add new information, and this should be checked on unseen data.

3.5 Dashboard

Three charts tell the story of the class. Most students score between about 55 and 80 (chart 1). The students at risk have low quiz averages (chart 2, saved as *class_dashboard.html*). Kicukiro has the highest average score and Gasabo the lowest, so Gasabo is where to act first (chart 3).

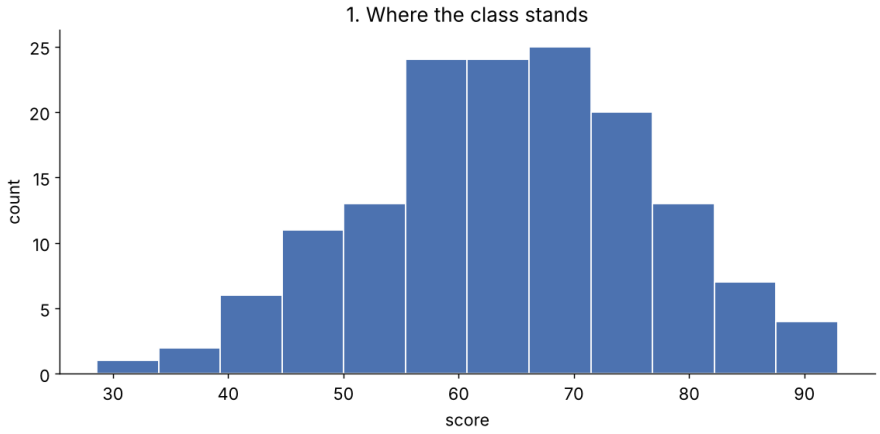


Chart 1. Where the class stands (histogram of scores, 12 bins)

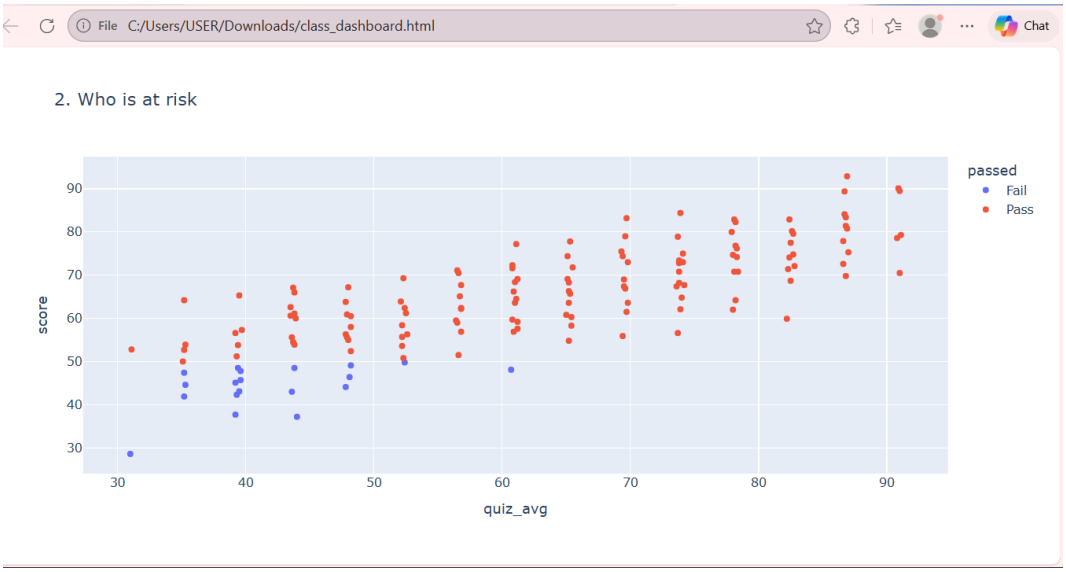


Chart 2. Who is at risk (class_dashboard.html opened in the browser)

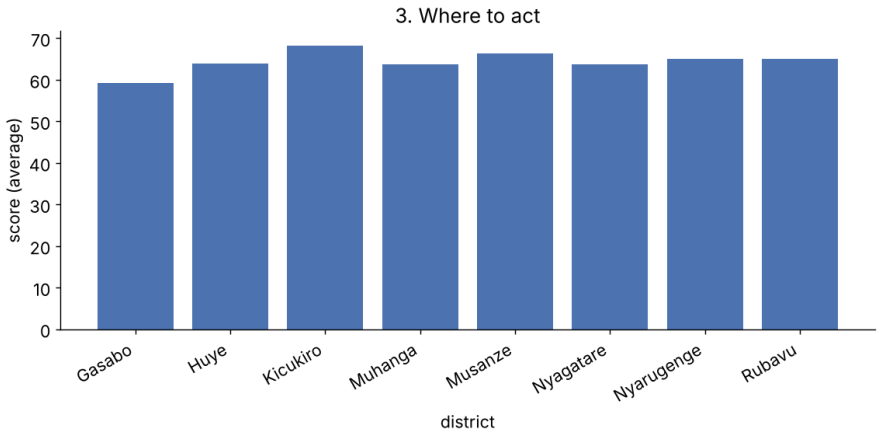
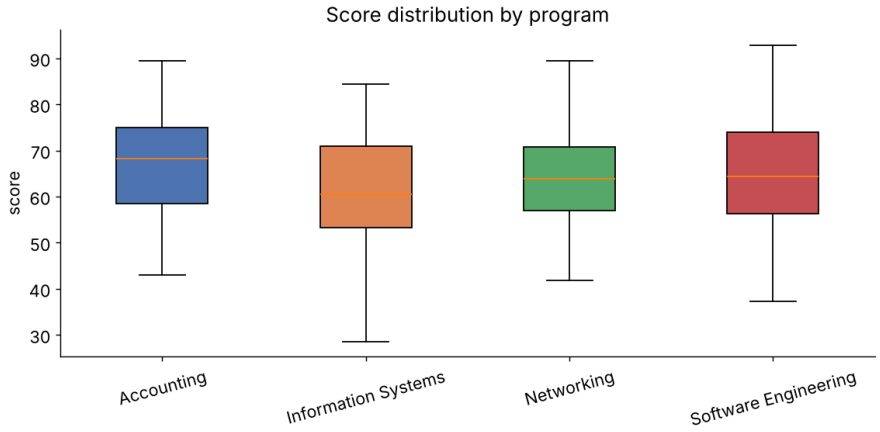


Chart 3. Where to act (average score per district)

3.6 Score distribution by program

Networking is the most consistent program (standard deviation 10.8), while Software Engineering has the widest spread (standard deviation 13.8, scores from 37.2 to 92.9).



Score distribution by program (box plot)

4. Model Card: Student Score Predictor

Section	Description
1. Purpose	Predicts a student's final score (0 to 100) from quiz average, assignments average and attendance. Meant for lecturers and the faculty to spot students who may need early support.
2. Data and features	150 AUCA students (week10_students.csv, cleaned). Inputs: quiz_avg, assignments_avg, attendance_pct. Target: score. Trained on 112 students, tested on 38 unseen students.
3. Performance	On unseen students the model is off by about 3.3 points on average (MAE = 3.33) and explains about 89% of the differences between scores (R^2 = 0.89).
4. Risks	A wrong prediction could label a capable student as at risk (stigma, lower expectations) or miss a student who needs help. The model only sees three numbers and does not know about illness, family problems, language or effort. **Use it to** start a conversation and offer support early. **Do not use it to** decide grades, pass/fail, scholarships or discipline, or to replace the lecturer's judgement. It was trained on 150 students from one course, so it may not work for other groups.

5. Conclusion

Quiz average, assignments average and attendance predict student scores well (MAE 3.33, R^2 0.89 on unseen students), and the model can flag most failing students early. The results also show that accuracy can be misleading on imbalanced data, and that extra features only help when they add new information. The predictions should support, not replace, the lecturer's judgement.